AN INVESTIGATION OF SKILL REQUIREMENTS FOR ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING JOB ADVERTISEMENTS

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ABSTRACT

Due to the advent of big data and efficient computational resources, Artificial Intelligence (AI) and Machine Learning (ML) have seen massive growth in recent years. Informatics degree programs are scrambling to meet the ever-increasing market demand of such professions. To explore the skillsets required for AI and ML positions, we conducted a content analysis of online job advertisements posted on Indeed.com. We present a ranking of the relevant skills for the two positions. Further, we performed a pairwise comparison between AI and ML positions. Overall, we observed that technical skills like data mining, programming, statistics, and big data are more valued for ML positions compared to AI positions. On the contrary, AI positions tend to be more generic, with an emphasis on communication skills. These clearly defined skills can be valuable for the hiring process as well as to revamp the existing course curricula to cater to the increasing market demand.

Keywords: Artificial Intelligence, Machine Learning, Skill Requirements, Content Analysis, Information Systems, Curriculum Design/Development

INTRODUCTION

According to the LinkedIn's 2020 Emerging Job Report (Romeo (2020)), Artificial Intelligence (AI) and Machine Learning (ML) jobs have been identified among the top emerging jobs of the year. The demand for these job positions has grown 74% annually in the last four years, made possible due to rapid advancement in new technologies and big data. The same findings are resonated by a Gartner Study (Meulen and Petty (2017)) which forecasts massive disruption of AI/ML in all industries (Wang and Siau (2019)). Both studies argue that the demand for AI/ML jobs is widespread. Despite the high demand, there is a shortage of skilled talent, which not only exists in the Information Technology (IT) industry but also in other traditional fields such as marketing, finance, healthcare, and supply chain (Wilson et al. (2017)). Thus, hire needs are spread across many industries, with most jobs requiring some knowledge of AI/ML technologies. Due to the interchangeable use of AI and ML for job advertisement and hire, there is a need for better stratification of the job skills between the two to make the hiring process more defined and efficient. Hence, one of the purposes of this research is to establish the similarities and differences among the two using the developed skill classification framework.

Recruiting for these AI/ML jobs in the industry can be extremely tough without training potential recruits early on in an academic setting. More specifically, new training modules are needed to prepare skilled employees for a competitive marketplace (Romeo (2020)). Skill development and technology adoption typically begin at the college or university level. With this realization, an overall increase in the number of analytics programs offering AI/ML skill training is becoming apparent. Due to ever-changing technology standards, there is a need for universities to continuously adapt their existing curricula to stay relevant for the skill requirements of these jobs.

There is also a growing need to clarify the definitions of job categories related to AI/ML. The current study attempts to address this by utilizing the associated skill categories. For this purpose, we examine the similarities and differences between AI and ML jobs with respect to different skills. A content analysis technique was used to extract keywords from online job postings to determine the frequencies of specific skills corresponding to each skill category. Using this, we identified the most critical skills as well as the associated skill category in our data sample and were able to assign relative importance to each skill for AI/ML jobs. The findings can also assist in curriculum design such that the most valuable skills are emphasized in the existing analytics curricula. Furthermore, the specific AI/ML courses can be updated to reflect the current in-demand software tools and applications.

The job positions that are part of this study represent professional areas of AI and ML. As discussed earlier, these two domains are closely interrelated. At present, the two terms are ambiguously defined and often used interchangeably in the literature. According to Goskel and Bozkurt (2019); Woschank et al. (2020); Wang and Siau (2019), the AI field deals with the development of technology capable of performing tasks related to human intelligence. More specifically, AI is the field of computer science dedicated to the creation of intelligent systems that try to mimic human behavior based on stipulated rules and algorithms (Jakhar and Kaur (2020)). ML and Deep Learning (DL) methods are integral parts of AI. This framework is summarized in Figure 1.

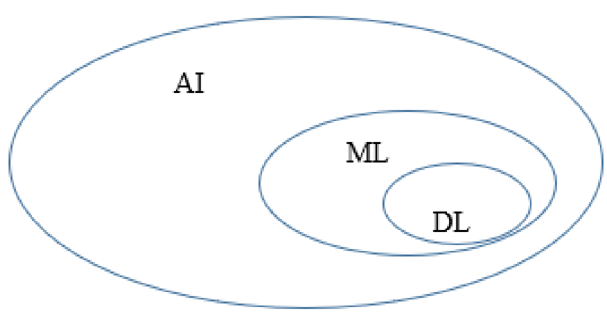

Figure 1. Hierarchy Of Different Domains

Machine Learning, although used interchangeably with AI, is a branch of AI that uses automated learning without any human intervention for a better understanding of data (Woschank et al. (2020); Agrawal, Jans and Goldfarb (2019)). ML models are usually applied to large datasets for pattern recognition, classification, and prediction tasks (Mitić, V. (2019)). This is usually achieved by minimizing an error or loss function. Depending on whether the output labels are known or unknown in advance, ML is further classified as Supervised or Unsupervised ML, respectively. DL is a subset of ML that involves higher accuracy models requiring computationally intensive resources like Graphical Processing Units (Jakhar and Kaur (2020)). The algorithmic components for DL are typically neural networks with important application areas like natural language processing, speech recognition, image recognition, recommender systems, etc. We would also like to emphasize that Data/Business Analytics is different from AI. In Data Analytics projects, the focus is mostly on descriptive statistics based on historical data (see Verma et. al. (2019) for more details). Therefore, this stream is also referred to as Descriptive Analytics. Sometimes, we could also use past data to predict what might happen in the future. These predictions rely on human interventions to understand data, identify trends, and create and test hypotheses. This stream is called Prescriptive Analytics. On the other hand, AI/ML systems are self-reliant and typically involve minimal supervision.

The interest in AI and ML technologies has been growing since the last decade. The universities would be benefitted from an in-depth study of the competency requirements in these fields. In this context, our aim is to answer the following research questions: (1) Which skills are required for AI and ML professionals? (2) What are the similarities and differences between the two job positions?

In the present study, we analyzed job advertisements across the US and provided a skill breakdown for the AI/ML positions. Using this perspective, we offer an alternate point of view of the employers' expectations. Using the sorted skill list based on relative frequency, educators can reconfigure the existing informatics programs so that emphasis is on the skillsets required in the job market. However, we do not provide specific recommendations in terms of curriculum design and reserve that topic for future research.

We propose to study only the "demand side" of the US job market by analyzing the job advertisements posted by employers. Future research could consider the gaps in the "supply side" by studying relevant university programs. This could entail a study of the course descriptions or course syllabi. Moreover, this could also be achieved by a survey of administrators and faculty members who are responsible for the design of such programs. Another research opportunity could explore the localization effects and study the differences in these requirements across multiple countries.

Our main contributions are the following: (1) We highlight the key technologies needed for the AI and ML positions along with summarizing the similarities and differences between the two. This is based on our competence framework that presents a broad classification of required skills sought by the employers in the two areas. (2) Our overview can provide guidance to the human resources team for the creation of job advertisements. This could also lead to the design of specialized in-demand training modules for the working professionals in the industry. (3) Educational institutions can incorporate our findings into their curricula. The results could be utilized to create new courses or redesign existing course offerings. Furthermore, the students interested in a related career could highlight these in-demand skills in their job applications.

This study supplements the existing literature of curriculum design in informatics programs by identifying key in-demand skills for two widely popular job categories. In addition to assisting academic institutions better align their coursework with the demands of the jobs, the results of our study could also be utilized by employers seeking highly skilled workers in AI/ML. It could also be beneficial in curating more consistent and better-defined job positions, highlighting the needed skillsets.

Our paper is organized as follows. First, we will discuss the impact of AI/ML on different disciplines. Next, we will examine the stream in the literature dealing with employers' perspective of job requirements. We will also investigate the pedagogical perspective, which provides a link between job requirements and degree programs. The classification framework, research method, and dataset will be discussed next. This will be followed by a discussion of the results focusing on each skill's relative importance corresponding to a specific job position. Additionally, we will examine the similarities and dissimilarities between AI and ML positions. Finally, we will outline the conclusions and directions for future research.

LITERATURE REVIEW

The role of AI/ML has been paramount in technological development leading to enhanced industry output (see Lu (2019) for a complete survey on AI models and application areas). Vochozka et al. (2018) estimated the economic impact of AI-based solutions on GDP to be around 24 billion dollars by 2035. The authors also document the estimated number of AI use cases, job impact, and the global merger and acquisition activity. The authors in Wamba-Taguimdje et al. (2020) studied the influence of AI on firm performance. The results highlighted that AI could benefit processes in finance, marketing, and administration. They also found that proper implementation of AI technology requires reconfiguration of the existing business process.

The impact of AI on education has also been well studied. Researchers have identified both benefits and problems associated with the use of AI. For example, McArthur et al. (2005) summarized the current AI applications in education, such as the Intelligent Tutoring System, and predicted its future use cases. The authors emphasized that a wide-scale implementation of AI in the classroom could be problematic. On the contrary, Roll and Wylie (2016) captured the evolution of AI technology adoption in the classroom through a bibliographic analysis, and argued that integrating AI into the learning environment could facilitate performing various tasks and achieving process-oriented goals. Along the same lines, Chassignol et al. (2018) identified four different streams of applications: educational content, innovative teaching method, enhanced assessment tool, and improved communication channel. More recently, Shar (2019) studied the students' perspective of AI adoption in education through a survey-based study and found both positive and negative impacts of AI. The authors established that variables like age and gender influence individual performance and adoption of AI.

Over the last couple of decades, several attempts have been made to integrate AI in the existing academic curriculum. For instance, by utilizing a computer-based modeling software, Schlanberger (1989) provided a framework for integrating the AI applications and methodology into the MBA program. Schlanberger (1989) was one of the first studies that explored the integration of AI into the business curriculum. The author proposes a framework for information systems planning based on four stages of the growth model, which can be utilized for curriculum design and resource planning. The lifecycle of an information system adoption is divided into four stages of initiation, expansion, formalization, and integration. They also highlight the support needed as the university passes through each stage of the life cycle along with examples of AI application development software and course syllabi.

Similarly, the importance of AI foundations in mathematics, science, engineering, and computer science curriculum was put forth by Aiken (1991), who also emphasized the value of laboratory component or hands-on experience with the current AI tools. Holder and Cook (2001) implemented this framework in the classroom using a web-based multimedia tool to develop and test AI algorithms. This simulation tool drew a lot of interest and elevated confidence of students in the class content. Similarly, Baldwin-Morgan (1995) noted that AI topics should be integrated into the accounting curriculum, and students should be exposed to AI tools before encountering them in the workplace. The author argued that students should be motivated with examples using expert systems and neural networks in auditing, taxation, cost accounting, management accounting, and financial accounting. Testa et al. (2001) designed a tool called Emergent-Design to integrate AI concepts in the architecture curriculum. This allowed students to learn about diverse roles in a design team while getting exposure to complex and dynamic design scenarios. More recently, Chedrawi and Howayeck (2019) argued that the fourth industrial revolution or Industry 4.0 is existent because of the technological advancements of AI. They proposed a model to integrate AI within the AACSB accreditation program through expert systems. This is expected to create a reliable and efficient process minimizing time, costs, and errors.

The impact of Industry 4.0 on jobs is also recognized by Romeo (2020). The author noted that rapid advancements in AI have dramatically changed the hiring landscape. The shortage of skilled AI talent has forced businesses to look for innovative solutions like upskilling. In this practice, the firms retrain their existing employees to develop expertise in AI. The influence of Industry 4.0 on job profiles and skills was also explored in Fareri et al. (2020) using text mining. They analyzed job descriptions from a multinational company, Whirlpool, with jobs and skills classified according to a dictionary of technology and methods. The authors provided measures to estimate the Industry 4.0 readiness of employees. Pejic Bach et al. (2020) also studied Industry 4.0 job descriptions using text mining to develop a job profile that highlighted the required skills. The first group of these job profiles focused on the Internet of Things and smart production design and control. The second group was more generic with application areas like supply chain, customer service, and enterprise software.

In addition to Industry 4.0, other related fields of interest are business analytics, data analytics, and big data. Numerous studies have explored the skill requirements in these professions using job advertisements. For example, Debortoli et al. (2014) studied the difference between big data and business intelligence job advertisements from an online employment portal. Their approach developed a competency framework based on a Latent Semantic Analysis of the text descriptions. Their results found that the business skills were as important as the technical skills for both the job positions. Gardiner et al. (2018) analyzed 1216 job advertisements, including “big data” in their title. Their research built on a framework of skill categories found that analytical skills and soft skills are highly demanded in addition to traditional hard skills. Lovaglio et al. (2018) web scraped Information and Communication Technology (ICT) and Statistical Italian job advertisements between June and September 2005. Their paper aimed to define these job positions in terms of the required skills. They also wanted to differentiate the Statistical positions from ICT positions. The results indicated that statisticians require more analytical skills and computing skills compared to soft skills. Verma et al. (2019) provided a skill breakdown for related professions such as Business Analyst, Business Intelligence Analyst, Data Analyst, and Data Scientist using content analysis of job advertisements. The authors concluded that the skills essential for these positions are decision making, organization, communication, and data management. More recently, Anton et al. (2020) studied the competencies required for three occupation fields: Data Science and Engineering, Software Engineering and Development, and Business Development and Sales. Their mixed methods approach combined a content analysis of scientific literature with a text mining of job advertisements. They presented the critical managerial and technical skills required for these positions.

The current study contributes to the existing literature by categorizing the AI and ML jobs in the analytics domain. AI and ML jobs have received limited attention, and therefore an in-depth analysis of the required skills is warranted. Also, we conduct a pairwise analysis of AI and ML jobs and discuss the similarities and differences between the two. Thus, this study aims to shed light on a significant job category in the Industry 4.0 era. Lastly, the current study contributes to the existing literature by introducing a novel set of skill categories using a skill classification framework for these two positions.

RESEARCH METHOD

Classification Framework

In this section, we describe how we developed a classification framework that consists of skill categories and a related set of skills. These skills were mapped against each job position. We used a comprehensive list of skills for various business and data analytics positions provided in Verma et al. (2019). We added more skills categories with associated skills to reflect the current market requirements for AI/ML positions. For this purpose, we manually investigated 100 job advertisements for each type. We asked three independent raters to validate our classification framework by placing skills into appropriate skill categories. The initial inter-coder reliability based on the alpha coefficient was observed to be 0.88. The skills and associated skill categories, along with sample keywords, are presented in Table 1.

Table 1. Classification Framework

Skill Category	Skills	Keywords
Communication	Written	Copywriting, Editing, Blogging, Content Creation, Story-ideation
	Verbal	Verbal, Oral, Cold calling
	Presentation	Present, Presentation, Report
	Generic	Responsible, Determined, Competitive, Witty, Success-oriented
Employee Attributes	Motivation	Motivated, Ambition, Willingness to learn, Delivering result, Continuous learning
	Time Management	Time management, Timely manner, Prioritize time, deadline driven
	Detail oriented	Attention to detail, Eye for detail, Accuracy, Precision
	Attitude	Can do, Go-getter, Self-learner, Self-directed, Positive Attitude
	Independence	Independence, Without supervision, Autonomous
	Adaptability	Adaptable, Flexible, Multitasking
	Confidence	Confident, Decisive
	Other	Funny, Smiling, High energy, Reliable, Proactive
Occupational Attributes	Enterprise System	ERP, CRM, SCM, SAP, PeopleSoft, Oracle, Integration, SAAS
	Visualization	Visualization, Tableau, Lumira, Crystal Reports, d3, d3.js
	Programming	Mathematical programming, Scala, Python, C#, C++, VB, Excel Macros, PERL, C, Java, Visual Basic, VB.NET, VBA, COBOL, FORTRAN, S, SPLUS, BASH, Javascript, ASP.NET, JQUERY, JBOSS
	Project Management	Project management, PERT, CPM, PERT/CPM, change management, project budget, project documentation, PMP, Microsoft Project, Gannt Chart
	Modeling	Neural networks, linear programming, integer programming, goal programming, queuing, genetic algorithms, expert systems
	Scraping	Scraping, web scraping, crawling, web crawling
	Hardware	Hardware, architecture, devices, printer, storage, desktop, pc, server, workstation, mainframe, legacy, system architecture
	Networks	Internet, LAN, WAN, networking, cloud computing, client server, distributed computing, network security, ubiquitous computing, TCP/IP
	Statistics	Statistics, SPSS, SAS, Excel, Stata, MATLAB, probability, hypothesis testing, regression, pandas, scipy, sps, spotfire, scikits.learn, splunk, h2o, R, STATA, Statistical programming
	Data Mining	Classification, text mining, web mining, stream mining, knowledge discovery, anomaly detection, associations, outlier, classify, association, estimation, prediction, forecasting, machine learning, decision trees
	Structured Data	SQL, relational database, Oracle, SQL Server, DB2, relational DBMS, Microsoft Access, data model, data management, entity relationship, data warehouse, DBMS, transactional database, sql server, db2, Cassandra, mongo db, mysql, postgresql, oracle db

	Big Data	Big Data, Unstructured Data, Data Variety, Data Velocity, Data Volume, Hadoop, Hive, Pig, Spark, MapReduce, Presto, Mahoot, NoSQL, Spark, shark, oozie, zookeeper, flume
	Decision Making	Reporting, analysis, modeling, design, problem-solving, implementation, testing, analytical, strategic thinking
	MS Office	MS Office, MS PowerPoint, presentation, MS Word, communication, documentation
Interpersonal	Interpersonal	Team management, Collaboration, Cooperation, Networking, Client relationship
Problem Solving	Problem Solving	Problem solving, Troubleshoot, Conflict resolution, Solve issue, Critical thinker
	Creativity	Creative, Out of box, Storyteller
	Process Design	Design Process, Improve process, Continuous improvement, Operations management
Administrative	Administrative	Issue management, Posting schedule, Product launch, social calendar
Analytical	Analytical	Insight, Identify trend, Summarize finding, Analyze trend, Synthesize information, Draw conclusion, Propose solution, Google Analytics, ArcGIS, GIS, QGIS, Data Analytics, Business Analytics
Research	Research	Data gathering, Data collection, Data reporting, Monitor trend, Monitor performance
Numeracy	Numeracy	Financial Management, Bookkeeping, Accountancy
Foreign Language	Foreign Language	Spanish, French, German, Italian, Chinese

Technique

Our technique of data collection and extraction is summarized in Figure 2. We used data from a popular online job portal, Indeed.com. The collected data based on US job advertisements was analyzed using content analysis (Neuendorf (2016)). Content analysis is a qualitative method that enables the tasks of extracting relevant information by counting the number of occurrences of specific keywords. The relevant information was extracted from job description field. The sentences were split into individual keywords known as unigrams. The consecutive words form bigrams and trigrams based on the distance between the neighbors. As part of the preprocessing routine, extraneous stopwords were removed from the analysis. The remaining set of keywords was matched against a keywords list for each skill presented in Table 1. In this way, we were able to determine which skills were required for each job position. Aggregating this knowledge across multiple job positions of the same type (AI or ML), we were able to determine which skills were relatively more important.

Our process can be divided into the following steps:

1. Data Collection: Web scraping was used as a primary method of data collection using Python. We queried Indeed.com for titles, including “Artificial Intelligence” and “Machine Learning” through its open-source API, whose input arguments are search phrase, location, and time period. We focused on jobs within the US between July

and December 2019. The API output includes the job URL and other fields like job title, summary, and location. The job summary field is a very abstract summary of the job description. Since we are interested in the complete job description, we designed a utility in Python which downloaded these job URLs and stored them locally.

2. Data Parsing: An advantage of relying on data from Indeed.com is that their web pages are standardized. Thus, the relevant information regarding job description can always be found under a specific HTML tag. This job description field contains a complete discussion of the requirements set by the employer for a specific job position. Thus, skill-based information could be located here. Our parser in Python extracted the correct tag corresponding to the job description field. Finally, we stored all job descriptions for each job position corresponding to AI or ML type.

3. Data Preprocessing: The job description field of each job posting consists of multiple sentences which are further broken down into individual keywords. These keywords contain contextual information through unigrams, bigrams, and trigrams. Note that unigram represents each word separately, while bigram represents two consecutive words in a specific sentence. For this purpose, each sentence is broken down into individual words. Thereafter, stopwords like a, an, the, and, in, for, etc. are removed from the analysis. The remaining individual words are identified as unigrams. The neighboring unigrams form bigrams and trigrams, depending on the distance between the neighbors. These n-grams were crucial to our analysis since the keywords belonging to each skill could be easily matched with the n-grams. If a match was found, we declared that a specific skill was required for a job advertisement.

4. Data Statistics: In this step, aggregate statistics regarding the skill requirements of AI/ML job positions were calculated. The n-grams from the previous step were compared against the keywords for each skill described in Table 1. Thus, we established which skills were demanded for each job advertisement. If a match was found, we conclude that a given job advertisement is linked to a corresponding skill of a specific skill category. This process was repeated for all job advertisements corresponding to AI/ML job type. For a specific job title, we collected the number of jobs associated with each skill. After that, we determined the relative frequency of occurrence of each skill by dividing with the total number of job postings of each type. We converted the relative frequency to a percentage count and presented it in Tables 4-5. This data will be utilized in the next section to classify skills concerning their relative importance for different job types.

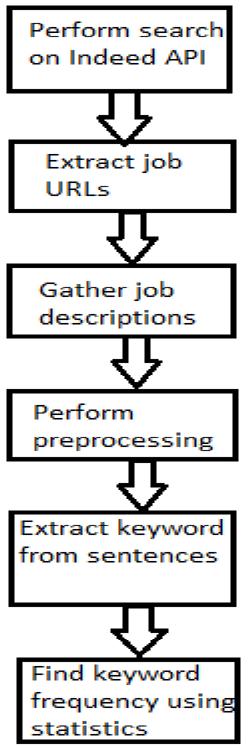

Figure 2. Summary Of Our Data Collection Routine

RESULTS

We queried Indeed.com for job titles containing phrases "Artificial Intelligence" and "Machine Learning" and obtained 1200 and 1700 entry-level jobs, respectively, between July and December 2019. The sample size is big enough for content analysis studies (Gardiner et al. (2018); Verma et al. (2019)). The geographical distribution in terms of the top five states of the respective jobs is shown in Table 2. As evident from the results, most of these jobs are located in CA, WA, and NY. The IT industry has seen a dramatic rise in the demand for AI/ML skilled workers, especially in the metropolitan areas of San Francisco, Seattle, Boston, and New York City.

Table 2. Geographical Distribution Of AI And ML Jobs

Rank	State (Percentage Count of AI jobs)	State (Percentage Count of ML jobs)
1	CA (34.50%)	CA (33.29%)
2	WA (10.83%)	NY (12.24%)
3	MA (7.00%)	MA (7.88%)
4	NY (6.83%)	WA (5.59%)
5	TX (4.92%)	VA (5.18%)

Next, we provide an in-depth analysis of the skill requirements for the AI and ML job categories. The five most common skill categories required for each job category are listed in Tables 4-5. These skill categories are sorted in descending order with respect to the number of job advertisements in which at least one skill associated with the skill category appears. For each skill category, we also report the top five skills. Again, these skills are ranked with respect to the percentage count associated with a specific skill relative to the total number of job postings for a specific job category.

To establish whether there is a significant difference between the two job types, we ran a hypothesis test. The percentage count of a specific skill with respect to AI and ML jobs facilitates a paired two-sample t-test. In this way, we compared the percentage relevance of each skill for AI against ML jobs. The null hypothesis was used to determine whether the mean difference between the two samples of observations is zero. We observed a p-value of 0.043, which helped us conclude that a minor difference exists between AI and ML job categories with a significance level of 5%. This is explained by the fact that ML is usually considered a subset of AI. Hence, many skill requirements will be the same for AI and ML jobs.

We also extracted the majors required for the job advertisements using regular expressions, which enables pattern matching in strings. For this purpose, we use the sentences in the job description of each job posting. We searched for keywords in the vicinity of major identifying information like B.S., B.A, Bachelor's, etc. These keywords were related majors like Artificial Intelligence, Machine Learning, Data Science, Computer Science, Information Technology, etc. The results are presented in Table 3.

Table 3. Majors Required For AI / ML Jobs

Rank	Major (Percentage Count of AI jobs)	Major (Percentage Count of ML jobs)
1	Machine Learning (70.00%)	Machine Learning (87.76%)
2	Engineering (68.83%)	Engineering (67.88%)
3	Business (61.25%)	Computer Science (59.59%)
4	Computer Science (59.17%)	Business (48.00%)
5	Artificial Intelligence (47.00%)	Mathematics (42.65%)
6	Mathematics (37.92%)	Data Science (28.82%)
7	Data Science (28.75%)	Statistics (26.94%)

Note that the percentage columns do not add up to 100% since one job posting could demand multiple majors like Bachelor's in Data Science or Engineering. Traditional degree programs in engineering, mathematics, and computer science are still demanded for AI/ML jobs. However, the newer niches like Data Science, Artificial Intelligence, and Machine Learning have started becoming more popular. We could observe that business degrees are also required since many of these positions deal with domain expertise. This is more pronounced for AI positions since ML positions tend to be slightly more technical.

Table 4. Artificial Intelligence Results

Skill Category	Skill	Percentage Count (%)
Occupation		
	Decision Making	87.42
	Data Mining	70.50
	Programming	65.58
	Statistics	37.58
	Big Data	32.75
Employee		
	Attitude	75.75
	Time Management	64.25
	Motivation	44.33
	Independence	36.08
	Attention to Detail	23.00
Communication		
	General	73.83
	Verbal	58.92
	Written	30.75
	Presentation	19.83
Interpersonal		
	Team	47.25
	Personal	13.17
	Networking	12.75
Problem Solving		
	General	32.50
	Creativity	29.17
	Process Design	28.75

Table 5. Machine Learning Results

Skill Category	Skill	Percentage Count (%)
Occupation		
	Data Mining	98.00
	Decision Making	86.88
	Programming	82.53
	Statistics	58.76
	Big Data	39.41
Employee		
	Attitude	98.00
	Time Management	52.71
	Motivation	41.71

	Independence	35.24
	Attention to Detail	18.47
Communication		
	General	61.24
	Verbal	40.35
	Written	33.47
	Presentation	17.76
Interpersonal		
	Team	44.82
	Personal	10.65
	Networking	5.24
Problem Solving		
	Process Design	24.88
	General	23.59
	Creativity	17.47

Next, we focus our attention on the similarities and differences between AI and ML positions. We compare the percentage count columns in Tables 4 and 5 and present an in-depth analysis of the required skills. The occupational skills are ranked at the top for both AI/ML positions. Decision-making skills are critical for both AI/ML positions. These include critical thinking skills while dealing with multiple data sources. The ability to synthesize information and draw conclusions to derive insights is essential in this data-driven digital economy. Knowledge of data mining techniques is more valued for ML positions. These consist of classical techniques like regression, classification, outlier detection, and forecasting with an increased focus on verticals like web mining, text mining, and stream mining. Statistical and big data skills also hold more significance for ML positions. The classical constructs of probability and hypothesis testing still hold some value. However, this field is dominated by various niche software packages like Numpy, Scikit-learn, Pandas, etc. along with traditional offerings like R, SPSS, SAS, etc. The advent of big data in terms of volume, variety, and velocity has necessitated new modeling paradigms. It has also generated an interest in software like Hadoop, Mapreduce, Nosql, etc.

The focus on hard skills like programming, data modeling, statistics, data mining, and structured data is more prominent for the ML positions. Moreover, for the ML job type, relatively more job postings require advanced knowledge of neural networks constructs like Convolution Neural Network, Deep Neural Network, Recurrent Neural Network (under the Modeling skill category). The programming skills in demand include Python and R in addition to the traditional languages like C, C++, and Java. These jobs also demand prior experience in design frameworks like Pytorch, Theano, Tensorflow, etc. Thus, the ML positions appear to be more technical. On the other hand, the overall systems design perspective through the knowledge of enterprise systems like Oracle, SAP, etc. is more highlighted for AI positions. Hence, AI job positions are more generic in the sense that holistic process understanding is critical for success.

Employee attributes have increasingly become more crucial in the workplace. Attitude is the most sought-after skill for both these positions. A can-do or go-getter attitude with a strong desire to learn is vital for success-oriented and ambitious professionals. Time management skills are especially important for millennials. Employees with a disposition towards setting and achieving priorities in a timely manner are valued in a fast-paced and dynamic business environment. The next key soft skills of motivation and independence are interrelated. An independent worker is a self-starter who completes tasks with minimal supervision. They are also proactive and take complete ownership of their projects. This requires them to be highly motivated, enthusiastic, and passionate. These traits are needed for success in any job. The next skill of attention to detail is considered a sign for organized and attentive individuals who can be an asset to any company. These findings, in terms of employee attributes, should be incorporated into the academic discourse, and the universities should adapt their curriculum accordingly.

The next important skill category is that of communication. The most common generic subtype includes keywords for responsible and determined employees who are ready to take initiatives and can deal with stress in a dynamic fast-paced environment. Note that this skill category is significantly more valued for AI positions. This is because most of the AI job requirements are generic and involve processing project requirements for systems-design framework development. This requires gathering various inputs from multiple team members and collating the results of the analysis. Hence verbal and written skills are more sought-after for AI positions than ML positions, which are somewhat technical. The presentation skills are also more vital for AI professionals since the project design phase typically involves meetings and brainstorming sessions with upper management.

The organization skills such as interpersonal skills are also critical for both job categories. They involve collaboration and cooperation in a team setting. The traits like leadership and networking with external clients and internal stakeholders are highly valued. The problem-solving skill category emphasizes individuals who are rational thinkers and can use sound judgement to troubleshoot and solve business issues. This is very useful in designing a business strategy for the long-term growth of a firm. The creativity or innovation skillset acknowledges the entrepreneurial spirit of the employee. Moreover, the new out-of-the-box ideas promote an employee’s intellectual development and generate new revenue channels for any organization. Lastly, the process design subtype of problem-solving signifies the relevance of the integrated processes at the firm level. The employees are expected to design and improve processes with an objective to minimize costs, time, and errors. Thus, knowledge of product design, operations management, supply chains, and marketing channels have become an important skill to acquire in the Industry 4.0. era.

DISCUSSION

Our research identified 35 skills organized into 10 different skill categories for AI and ML professions (presented in Table 1). While the literature has provided some insights related to AI (Anton et al. (2020); Balan (2019); Bawack et al. (2019)), our comprehensive analysis is centered on empirical data. More specifically, we explore the employer's perspective through analysis of the job advertisements.

Most of the recent interest in AI and ML technologies have resulted from the information technology sector. Hence, the jobs are typically found in the metropolitan hubs located in NY, WA, and CA. From Table 3, it is clear that the business-oriented majors are more in demand for AI professions. The business-related competencies provide domain level expertise in fields like healthcare and digital marketing. Moreover, we observe that statistics majors are more valued for the ML professions. This emphasizes that the required ML job skills are somewhat more technical with a special focus on data driven applications in banking, finance, and supply chain management.

We identified a number of similarities between AI and ML job postings. The business skills and generic IT concepts are common to these two positions. It is well established that all strategic decision-makers could benefit from a sound understanding of their business to increase the effectiveness of AI solutions (Brynjolfsson and Mitchell (2017); Mohanty and Vyas (2018)). The technical expertise in data analysis, visualization, and reporting skills are also highly valued. Tasks like classification and prediction are central to the areas of supervised and unsupervised learning. These skills are highly sought by employers in addition to outlier detection and forecasting techniques. The classical field of data mining concerns itself with pattern extraction on structured data. Due to the abundance of data in terms of volume, variety, and speed, the newer domains of web mining and stream mining has started to become more popular. They require expertise in both unstructured database management as well as some specific software programs. These jobs increasingly require decision-making by extracting information from web-based and unstructured data sources related to text mining, web mining, and social networks. We have seen from the results that these big data skills have started gaining prominence. This is evident from the focus on big data tools like

Hadoop, Mapreduce, Hive, etc. Moreover, skills in traditional relational databases like SQL and Oracle have declined in popularity.

We also note that there are some clear differences between AI skills and ML skills. The ML competency places more emphasis on computer programming skills. The newer programming languages like Python are slowly catching up to traditional languages like C++ and Java. These findings match the insight provided by the Google Trends data with respect to the volume of search queries (cf. Figure 3). Almost 60 percent of the ML jobs we analyzed asked for strong software programming skills and statistical knowledge, whereas AI jobs required much less "programming" and statistical knowledge. In addition, ML jobs demand data analysis and machine learning skills like outlier detection, regression, classification, etc. In comparison, there is less mention of such terms in AI jobs.

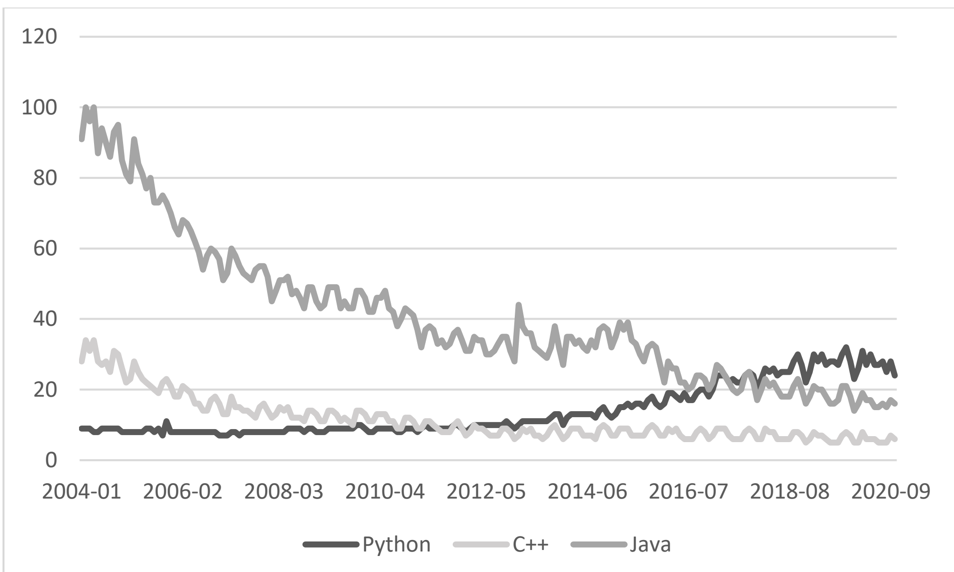

Figure 3: Google search volume in the US for the search queries of the different programming languages (Source: Google Trends)

We also want to mention that the current job listings demand more expertise in "open-source" tools. This reduces the "barriers of entry" for these professions. Even though large vendors like SAS, SPSS, SAP, etc. provide various analytical solutions in these domains, the job market does not place more emphasis on them.

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CONCLUSIONS AND FUTURE RESEARCH

In this research, we studied the job descriptions for the AI/ML positions in the US. We utilized content analysis to develop a ranking of key skill categories required for each job type. Moreover, we performed a pair-wise comparison of the top five skill categories for AI and ML. We found that occupational skills are most sought-after for the two job positions. ML positions are somewhat more technical and place more emphasis on data mining, programming, statistics, and big data skills. On the other hand, AI job positions assign more importance to the communication skills signaling that it requires interaction with other team members and upper management while working on a proof-of-concept for systems design projects.

Our current study identified and ranked skills that are currently required for AI/ML positions. Our findings could be used in two different ways. First, the Management Information Systems, Data Science, Data Analytics, and Business Analytics programs could revamp their existing offerings related to AI/ML. Thus, our study could be beneficial for designing associated undergraduate and graduate degree programs. Second, the findings could be used by hiring managers looking for skilled AI/ML workers. By using the proposed skills, the hiring team could create well-structured and consistent job descriptions. In this way, the hiring team could utilize their employees efficiently.

Future research could involve a localized study of some specific US states. This would benefit the local employers as well as local universities. Another research direction could be to investigate job descriptions from other professional fields like accounting, finance, marketing, and healthcare. In addition, scholars could also analyze the existing curricula of degree programs in the AI/ML domain. In this way, the findings of the present study could be used parallelly with the coursework of degree programs in the related fields. By identifying the skill gaps in the existing programs, we could make more concrete recommendations regarding curriculum design.

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